

EmerFlux: A Two-Layer Liquid Surface Display for Organic Pixel-Based Aesthetic Representation of Information

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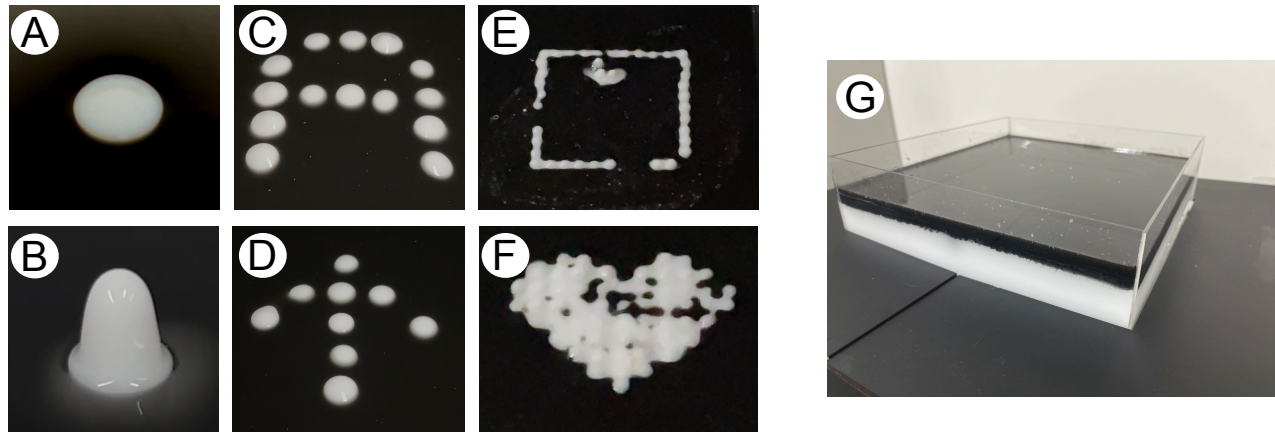


Figure 1: EmerFlux examples and system overview. (A) a flat circular pixel, (B) an actuated protruding pixel, (C) text rendering with pixels, (D) an arrow icon, (E) a line-like stroke formed by merging adjacent pixels, (F) a surface-like region formed by merged pixels, and (G) the tabletop liquid-surface display setup.

Abstract

This paper proposes EmerFlux, a surface-based liquid display that presents protruding pixels on a water surface using an immiscible two-layer liquid structure composed of water and oil. By ejecting water upward through an array of independently controlled pumps, the system generates pixel-like protrusions directly on the liquid interface.

EmerFlux enables information to be presented not only as discrete points, but also as connected line- and surface-like forms through interactions between neighboring pixels. Leveraging the dynamic behavior inherent to liquids, these expressions exhibit smooth temporal transitions and continuous morphological changes, highlighting the temporal texture and material expressiveness of liquid surfaces.

At SIGGRAPH Emerging Technologies, EmerFlux will be presented as a tabletop-scale liquid surface exhibit. Visitors can observe characters, icons, and dynamic patterns emerging directly on the

water surface, demonstrating new possibilities for information presentation through continuously transforming liquid media.

CCS Concepts

• Human-centered computing → Displays and imagers.

ACM Reference Format:

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1 Introduction

Water is an essential material for human life and has long been deeply connected to everyday living and culture. Changes in landscapes reflected on the surfaces of ponds and rivers have been appreciated, and fountains in urban spaces have also functioned as public art through their constantly transforming forms. Thus, water surfaces have been explored as expressive media due to their fluidity, reflection, and temporal transformation.

For example, Sakamoto's work *water state 1* treats the water surface as a material embodying environmental change and temporality [Sakamoto and Takatani 2013]. More recently, bubble-based displays [Ishii and Siio 2019] and fluid surface expressions using



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magnetic fluid [Morita 2025] have proposed methods for structuring unstable liquid phenomena into organic pixel-like units.

However, using water surfaces as a medium for information presentation requires sufficient visual contrast, stable control over the appearance and disappearance of displayed patterns, and rapid updates. Many prior approaches rely on particle movement, bubble generation, material state changes, or the actuation of rigid pixels, which limits their ability to control the liquid surface itself rapidly and with high contrast.

In this work, we propose EmerFlux, a liquid-surface display that uses an immiscible two-layer liquid structure composed of water and oil to generate white protruding pixels on a black liquid surface. EmerFlux forms stable white water protrusions on a black oil surface, enabling high-contrast pixel representations. Furthermore, an independently controlled 11×11 pump array enables rapid pixel switching and dynamic pattern presentation on the liquid surface. Through this approach, EmerFlux realizes information representations that incorporate fluid, temporally changing, and organic motion.

2 Implementation

EmerFlux is a liquid-surface display composed of an immiscible two-layer liquid interface and a pump-driven actuation system. The system locally injects water from the lower layer using water pumps to deform the water–oil interface, generating white protrusions directly on the surface. By treating these white protrusions as pixels, EmerFlux presents information on the liquid surface. Figure 2 illustrates the operating principle of EmerFlux, from the resting state to pixel appearance and disappearance.

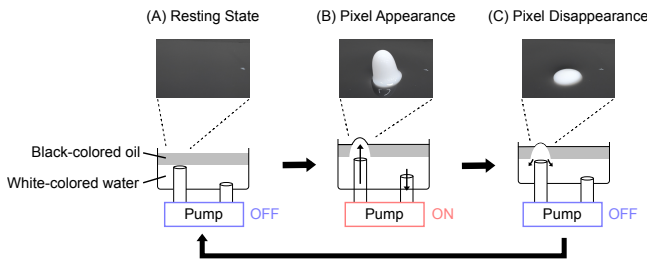


Figure 2: Operating principle of EmerFlux: (A) resting state, (B) pixel appearance, and (C) pixel disappearance.

The two-layer liquid structure consists of black-colored oil above white-colored water. As shown in Figure 2, when the pump is activated, white water emerges through the oil layer and forms a protruding pixel on the black oil surface, enabling high-contrast visual expression. When pumping stops, the protrusion naturally settles back beneath the oil layer, and the white pixel smoothly disappears.

3 Display Capability & Visitor Experience

EmerFlux is a liquid surface display that independently controls protruding pixels on a water surface using 121 submersible pumps arranged in an 11×11 grid. The system enables rapid on/off switching of pixels without requiring physical particle movement. In single-pixel experiments, individual protrusions could be activated and

deactivated within approximately 0.2–0.4 seconds. The generated pixels have a maximum height and diameter of 1.5 cm, providing sufficient visibility for tabletop-scale viewing.

Using this pixel array, EmerFlux can display recognizable patterns such as characters and icons directly on the water surface. Pixels emerge as white circular protrusions from a black liquid background and rise and settle in accordance with the inherent dynamics of fluid motion. This behavior produces smooth temporal transitions and emphasizes the material presence of the water surface.

By arranging the nozzles at 4 mm intervals, neighboring pixels can be connected in a controlled manner. As a result, the display supports continuous liquid expressions that extend beyond discrete point-based representations into line- and surface-like forms. By modulating injection volume and timing, information can appear on the water surface as points, strokes, or filled regions.

As an exhibit experience, visitors observe the water surface itself transforming into a dynamic display medium. Rather than simply switching pixels on and off, EmerFlux presents information through processes of emergence, connection, and dissolution. Through this approach, EmerFlux proposes a novel surface-based display platform in which computation and physical behavior coexist on the boundary of immiscible fluids.

4 Conclusion

EmerFlux is a surface-based liquid display that presents protruding pixels with clear contrast on a water surface using an immiscible two-layer liquid structure composed of water and oil. By combining independently controlled actuation with the dynamic behavior inherent to liquids, the system enables information to emerge on the water surface not only as discrete pixels, but also as connected line- and surface-like forms.

This approach extends pixel-based representation in physical displays beyond the control of discrete pixels. By using liquid as a continuous material, EmerFlux enables not only pixel appearance and disappearance or height variation, but also connections between neighboring pixels and organic transitions from emergence to disappearance. In this way, this work reconsiders pixels as connected and temporally changing expressive units on a continuous liquid surface, rather than merely as independent display units.

At SIGGRAPH Emerging Technologies, EmerFlux will be presented as a tabletop-scale liquid surface exhibit. Visitors will be able to observe characters, icons, and dynamic patterns emerging directly on the water surface. The demonstration presents both expressions based on discrete protruding pixels and continuous expressions that expand into line- and surface-like forms, showcasing the ability to switch between these modes.

Through these expressions, EmerFlux presents new possibilities for liquid-based displays in which information presentation is integrated with the material behavior of liquids.

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